

Question 1:

(10 marks)

Perform Signed Multiplication using Booth's algorithm? Where Multiplicand is 059H and Multiplier is (-21)₁₀?

A(Accumulator)	Q (Multiplier)	Q ₋₁	Steps	
0000 0000	11101011	0	initialize	
1010 0111	1110 1011	0	A = A - M	(8)
1101 0011	1111 0101	1	SAR	
1110 1001	1111 1010 6th	1	SAR	(7)
0100 0010	1111 1010	1	A = A + M	
0010 0001	0111 1101	0	SAR	(6)
1100 1000	0111 1101	0	A = A - M	
1110 0100	0011 1110	1	SAR	(5)
0011 1101	0011 1110	1	A = A + M	
0001 1110	1001 1111	0	SAR	(4)
1100 0101	1001 1111	0	A = A - M	
1110 0010	1100 1111	1	SAR	(3)
1111 0001	0110 0111	1	SAR	(2)
1111 1000	1011 0011	1	SAR	(1)

Question 2:

(10 marks)

Perform Unsigned Multiplication using Booth's algorithm? Where Multiplicand is and 0DAH
Multiplier is (113)₁₀?

C	Accumulator	Q	Steps	
0000 0000	0000 0000	0111 0001	initialize	
0000 0000 0000 0000	1101 1010 0110 1101	0111 0001 0011 1000	A+M SHR	(7)
0000 0000	0011 0110	1001 1100	SHR	(6)
0000 0000	0001 1011	0100 1110	SHR	(5)
0000 0000	0000 1101	1010 0111	SHR	(4)
0000 0000 0000 0000	1110 0111 0111 0011	1010 0111 1101 0011	A+M SHR	(3)
0000 0001 0000 0000	0100 1101 1010 0110	1101 0011 1110 1001	A+M SHR	(2)
0000 0001 0000 0000	1000 0000 1100 0000	1110 1001 0111 0100	A+M SHR	(1)
0000 0000	0110 0000	0011 1010	SHR	(0)

Question 3

(5 marks)

Run following code and update registers accordingly in HEXA?

clc	AX	B5C9
mov ax,0B5C9H	AL	24
SHL al,2	AX	f6a4
SAR ax,3	AX	ed48
SAL ax,1	AL	24
SHR al,1	AX	d24e
ROL ax,4	AL	13
RCR al,2	AX	a427
RCL ax,1	AX	27a4
ROR ax,8		